

Location: Boston Logan International Airport

Motivation:

Airports are particularly bad offenders in terms of carbon emissions, so reducing Logan's dependence on energy from fossil fuels could have a significant impact on the airport's environmental footprint. Plus, Mass Transportation is trying to be net zero by 2028, so this is something that the state cares about (and has invested a lot of money into).

<https://www.massport.com/environment/roadmap-to-net-zero>

Wind data: <https://mesonet.agron.iastate.edu/request/asos/1min.phtml>

collected at 9m from ground

Avg wind speed: 4.8 m/s

Avg gusts: 6.3 m/s

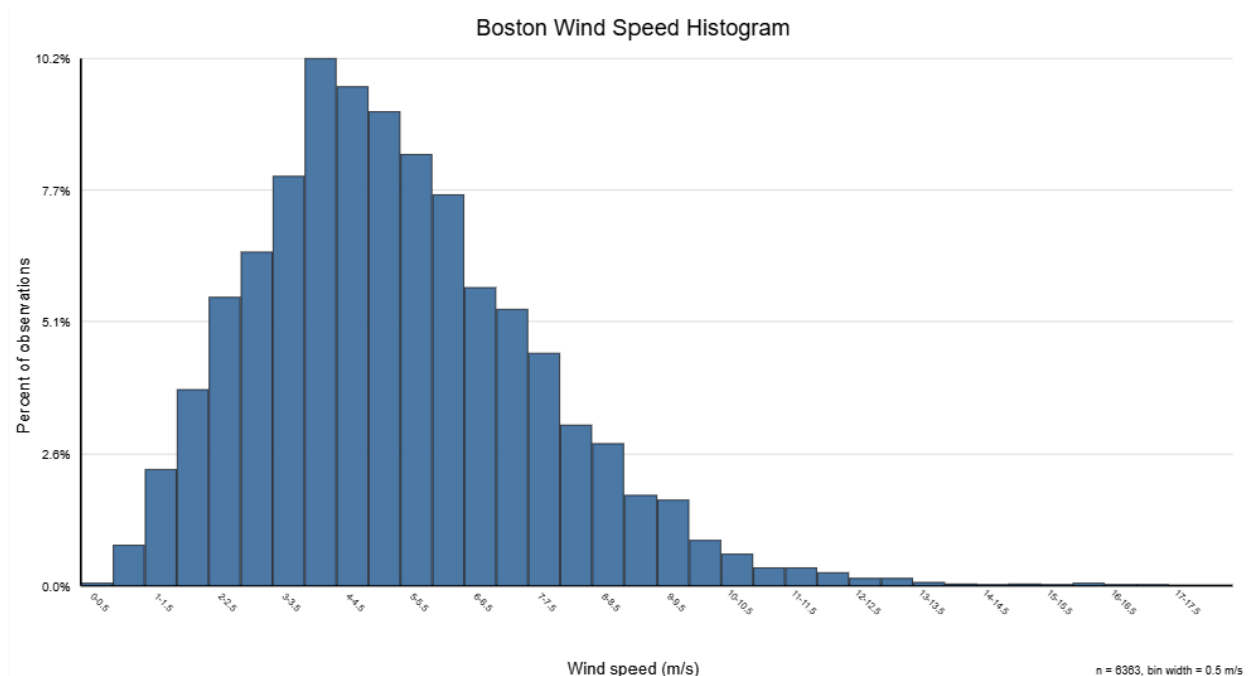


Figure 1: Histogram of hourly wind speed from Jul 2025 - July 2026. There are significant gaps so only about 73% of the year is represented. For AEO calculations we will use a 5 year span.

Obstacles:

- Need to figure out restrictions on placing these in near runway areas
 - The weather station where the wind data was collected seems to be a 30ft tall structure in a large field in between runways... can we do that?
 - How many turbines can we put there?
- Could we put turbines on the roof of the airport? What height will conflict with airspace?

- This is likely out of the scope of our project due to time constraints but, presumably there is some location near the airport, around 9m off the ground with similar wind conditions where the turbines could be placed, if not where we are suggesting.

Design thoughts:

- Most Darrieus turbines have rated speed around 10-15 m/s and cut-out speed around 20 m/s according to the LLM wiki
- Va9 EN0005 Darrieus turbine has a cut in speed < 2 m/s!
- should design turbine to be at a similar height as wind measurements for best estimations of power production. (Wind speed can vary dramatically in the turbulent boundary layer near the ground.)
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